Class 12 Mathematics

Chapter 1: Relations and Functions

Practice Set B: step-wise solutions

20 questions **52** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 **MCQ** NCERT Ex 1.1 Q15 — 1 mark

Let R be the relation in the set $\{1, 2, 3, 4\}$ given by $R = \{(1, 2), (2, 2), (1, 1), (4, 4), (1, 3), (3, 3), (3, 2)\}$. Then:

(A) R is reflexive and symmetric but not transitive

(B) R **is reflexive and transitive but not symmetric**

(C) R is symmetric and transitive but not reflexive

(D) R is an equivalence relation

Formula / idea: Transitive: whenever $(a, b) \in R$ and $(b, c) \in R$, also $(a, c) \in R$.

1	Reflexive: all of $(1, 1), (2, 2), (3, 3), (4, 4)$ are in R. $(a, a) \in R$ for every a	
2	Symmetric? $(1, 2) \in R$ but $(2, 1) \notin R$. $\Rightarrow$ R is not symmetric	
3	Transitive: check every chain $(a, b), (b, c)$ with $a \neq b \neq c$. $(1, 3), (3, 2) \in R \Rightarrow (1, 2) \in R$ ✓	
4	The other chains end in a pair (b, b), so they give nothing new. R is transitive.	✓ 1
5	Reflexive and transitive but not symmetric. Option (B).	

Answer **(B) R is reflexive and transitive but not symmetric**

Watch out: Missing the chain $(1, 3), (3, 2)$ and calling R not transitive — $(1, 2)$ is present.

2 MCQ NCERT Misc. Ex. Q6 1 mark

Let $A = \{1, 2, 3\}$. Then the number of relations containing $(1, 2)$ and $(1, 3)$ which are reflexive and symmetric but not transitive is:

(A) 1 **(B)** 2 **(C)** 3 **(D)** 4

Formula / idea: Build the smallest relation with the required properties, then see what else can be added.

1 Reflexive: add all (a, a). Symmetric: add $(2, 1)$ and $(3, 1)$.
$R_1 = \{(1,1), (2,2), (3,3), (1,2), (2,1), (1,3), (3,1)\}$

2 R_1 is not transitive.
$(2, 1), (1, 3) \in R_1$ but $(2, 3) \notin R_1$

3 The only pairs left are $(2, 3)$ and $(3, 2)$; symmetry forces both together.
$R_1 \cup \{(2,3), (3,2)\} = A \times A,$ which is transitive

4 So R_1 is the only such relation. ✓ 1
Number of relations $= 1$

5 Option (A).

Answer **(A)** 1

Watch out: Adding $(2, 3)$ alone — symmetry then forces $(3, 2)$, giving $A \times A$, which is transitive.

3 MCQ PYQ-style 1 mark

Let $f : \mathbb{R}_+ \to [-5, \infty)$ be defined as $f(x) = 9x^2 + 6x - 5$, where $\mathbb{R}_+$ is the set of all non-negative real numbers. Then f is:

(A) one-one but not onto **(B)** onto but not one-one
(C) **bijective** **(D)** neither one-one nor onto

Formula / idea: Complete the square: $9x^2 + 6x - 5 = (3x + 1)^2 - 6$.

1 Complete the square.
$f(x) = (3x + 1)^2 - 6$

2 One-one: $f(x_1) = f(x_2)$ with $x_1, x_2 \geq 0$.
$(3x_1 + 1)^2 = (3x_2 + 1)^2 \Rightarrow 3x_1 + 1 = 3x_2 + 1 \Rightarrow x_1 = x_2$
both $3x_1 + 1$ and $3x_2 + 1$ are positive

3 Onto: for $y \geq -5$, solve $f(x) = y$. ✓ 1
$x = \dfrac{\sqrt{y+6} - 1}{3} \geq 0 \quad (\text{as } y + 6 \geq 1)$

4 One-one and onto, so f is bijective. Option (C).

Answer (C) bijective

Watch out: Calling it many-one like x^2 on $\mathbb{R}$ — on $x \geq 0$ the bracket $3x+1$ never changes sign.

4 MCQ CBSE 2024 (65/3/3) Q8 1 mark

Let $\mathbb{Z}$ denote the set of integers, then function $f : \mathbb{Z} \to \mathbb{Z}$ defined as $f(x) = x^3 - 1$ is:

(A) both one-one and onto **(B)** **one-one but not onto**

(C) onto but not one-one **(D)** neither one-one nor onto

Formula / idea: Onto needs a pre-image **inside the domain** $\mathbb{Z}$ for every integer in the codomain.

1 One-one: let $f(x_1) = f(x_2)$.
$x_1^3 - 1 = x_2^3 - 1 \Rightarrow x_1^3 = x_2^3 \Rightarrow x_1 = x_2$

2 Onto? Take $y = 1$ in the codomain. ✓ 1
$x^3 - 1 = 1 \Rightarrow x = \sqrt[3]{2} \notin \mathbb{Z}$

3 So 1 has no pre-image in $\mathbb{Z}$: f is not onto.

4 One-one but not onto. Option (B).

Answer (B) one-one but not onto

Watch out: Thinking 'cubes are onto' — that is true on $\mathbb{R}$, not on $\mathbb{Z}$.

5 MCQ NCERT Exemplar Ex 1.3 Q35 1 mark

If the set A contains 5 elements and the set B contains 6 elements, then the number of one-one and onto mappings from A to B is:

(A) 720 **(B)** 120

(C) 0 **(D)** none of these

Formula / idea: A one-one map from an m-element set has exactly m images.

1 A one-one map from A uses exactly 5 different images.
$n(\text{range}) = 5$

2 But onto needs all 6 elements of B as images. ✓ 1
$5 < 6 \Rightarrow$ not onto

3 So the number of one-one and onto maps is 0. Option (C).

Answer **(C)** 0

Watch out: Answering ${}^6P_5 = 720$ — that counts one-one maps, none of which is onto.

6 **Assertion-Reason** Board-pattern 1 mark

Assertion (A): The relation $R = \{(a, b) : ab > 0\}$ on the set $\mathbb{R}$ of real numbers is symmetric and transitive but not reflexive. **Reason (R):** Every relation which is symmetric and transitive is also reflexive.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) **Assertion (A) is true, but Reason (R) is false.**

(D) Assertion (A) is false, but Reason (R) is true.

Formula / idea: Symmetric and transitive give (a, a) only for those a that are related to some b.

1 Check A: R is symmetric and transitive.
$ab > 0 \Rightarrow ba > 0; \qquad ab > 0,\ bc > 0 \Rightarrow ab^2c > 0 \Rightarrow ac > 0$
$b \neq 0$, so $b^2 > 0$

2 But R is not reflexive, so A is true.
$0 \cdot 0 = 0 \not> 0 \ \Rightarrow \ (0, 0) \notin R$

3 Check R: the relation in A itself is symmetric and transitive but not reflexive. ✓ 1
So R is false

4 A false statement cannot explain A.

5 A is true, R is false. Option (C).

Answer **(C) A is true, but R is false**

Watch out: 'Proving' reflexivity from $(a, b), (b, a) \in R$ — it fails for $a = 0$, which is related to nothing.

7 **Short answer** Board-pattern 2 marks

Show that the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 4x^3 + 5$ is both one-one and onto.

Formula / idea: Every real number has exactly one real cube root.

1 One-one: let $f(x_1) = f(x_2)$.
$4x_1^3 + 5 = 4x_2^3 + 5 \ \Rightarrow \ x_1^3 = x_2^3$

2	Take cube roots. $x_1 = x_2 \Rightarrow f$ is one-one	✓ 1
3	Onto: let $y \in \mathbb{R}$ and solve $f(x) = y$. $x = \sqrt[3]{\dfrac{y-5}{4}} \in \mathbb{R}$ a cube root of any real number is real	✓ ½
4	Check the pre-image. $f(x) = 4 \cdot \dfrac{y-5}{4} + 5 = y$	✓ ½
5	Every y has a pre-image. Hence f is one-one and onto.	

Answer f **is one-one and onto (bijective).**

Watch out: Writing $x_1^3 = x_2^3 \Rightarrow x_1 = \pm x_2$ — that is for squares, not cubes.

Level 2

19 marks

8 **Short answer** Board-pattern 2 marks

Show that the relation R on the set $\mathbb{Z}$ of integers, given by $R = \{(a, b) : a^2 + b^2 \text{ is divisible by } 2\}$, is an equivalence relation.

Formula / idea: An equivalence relation is reflexive, symmetric and transitive.

1	Reflexive: put $b = a$. $a^2 + a^2 = 2a^2$, divisible by 2 $\Rightarrow$ $(a, a) \in R$	✓ ½
2	Symmetric: let $(a, b) \in R$. $b^2 + a^2 = a^2 + b^2$, divisible by 2 $\Rightarrow$ $(b, a) \in R$	✓ ½
3	Transitive: let $a^2 + b^2 = 2k$ and $b^2 + c^2 = 2m$. Add them. $a^2 + 2b^2 + c^2 = 2(k + m)$	
4	Rearrange. $a^2 + c^2 = 2(k + m - b^2) \Rightarrow (a, c) \in R$	✓ ½
5	Hence R is an equivalence relation.	✓ ½

Answer R **is reflexive, symmetric and transitive, hence an equivalence relation.**

Watch out: Writing $a^2 + c^2 = 2(k + m)$ — the $2b^2$ term must be taken across first.

9 **Short answer** CBSE 2024 (65/4/1) Q33(b) 3 marks

Check whether the relation S in the set of real numbers $\mathbb{R}$ defined by $S = \{(a, b) : a - b + \sqrt{2} \text{ is an irrational number}\}$ is reflexive, symmetric or transitive.

Formula / idea: One counter-example is enough to disprove a property.

1	Reflexive: for every $a \in \mathbb{R}$. $a - a + \sqrt{2} = \sqrt{2},\ \text{irrational} \Rightarrow (a, a) \in S$	✓ 1
2	Symmetric? Take $a = \sqrt{2}, b = 0$. $\sqrt{2} - 0 + \sqrt{2} = 2\sqrt{2}\ (\text{irrational}) \Rightarrow (\sqrt{2}, 0) \in S$	
3	Reverse the pair. $0 - \sqrt{2} + \sqrt{2} = 0\ (\text{rational}) \Rightarrow (0, \sqrt{2}) \notin S$	✓ 1
4	So S is not symmetric.	
5	Transitive? Take $a = \sqrt{2}, b = 1, c = 2\sqrt{2}$. $a - b + \sqrt{2} = 2\sqrt{2} - 1, \quad b - c + \sqrt{2} = 1 - \sqrt{2}\quad (\text{both irrational})$	
6	So $(a, b), (b, c) \in S$. But: $a - c + \sqrt{2} = \sqrt{2} - 2\sqrt{2} + \sqrt{2} = 0 \Rightarrow (a, c) \notin S$	✓ 1
7	Hence S is reflexive, but neither symmetric nor transitive.	

Answer S **is reflexive, but neither symmetric nor transitive.**

Watch out: Testing only rational numbers — every counter-example here needs $\sqrt{2}$ to cancel.

10 **Short answer** Board-pattern 3 marks

Let $A = \{1, 2, 3, \ldots, 12\}$ and R be the relation in $A \times A$ defined by $(a, b)\, R\, (c, d)$ if $ab = cd$. Prove that R is an equivalence relation and obtain the equivalence class $[(3, 4)]$.

Formula / idea: $[(a, b)]$ is the set of all pairs in $A \times A$ related to (a, b).

1	Reflexive: for any $(a, b) \in A \times A$. $ab = ab \Rightarrow (a, b)\, R\, (a, b)$	✓ ½
2	Symmetric: let $(a, b)\, R\, (c, d)$. $ab = cd \Rightarrow cd = ab \Rightarrow (c, d)\, R\, (a, b)$	✓ ½
3	Transitive: let $(a, b)\, R\, (c, d)$ and $(c, d)\, R\, (e, f)$. $ab = cd = ef \Rightarrow (a, b)\, R\, (e, f)$	✓ ½
4	Hence R is an equivalence relation.	✓ ½

5 $[(3,4)]$ has every pair in $A \times A$ with product $3 \times 4 = 12$.
$[(3,4)] = \{(a,b) \in A \times A : ab = 12\}$

6 List the factor pairs of 12 with both factors in A. ✓ 1
$[(3,4)] = \{(1,12),(2,6),(3,4),(4,3),(6,2),(12,1)\}$

Answer R **is an equivalence relation;** $[(3,4)] = \{(1,12),(2,6),(3,4),(4,3),(6,2),(12,1)\}$

Watch out: Forgetting $(1,12)$ and $(12,1)$ — $12 \in A$, so they belong to the class.

11 **Short answer** Board-pattern 3 marks

Let $f : \mathbb{N} \to \mathbb{Z}$ be defined by $f(n) = \begin{cases} \frac{n-1}{2}, & \text{if } n \text{ is odd} \\ -\frac{n}{2}, & \text{if } n \text{ is even} \end{cases}$. Show that f is both one-one and onto.

Formula / idea: Odd n give images ≥ 0; even n give images < 0.

1 Note the signs of the images.
n odd $\Rightarrow f(n) = \frac{n-1}{2} \geq 0, \qquad n$ even $\Rightarrow f(n) = -\frac{n}{2} < 0$

2 One-one: let $f(n_1) = f(n_2)$. Both odd: ✓ ½
$\frac{n_1 - 1}{2} = \frac{n_2 - 1}{2} \Rightarrow n_1 = n_2$

3 Both even: ✓ ½
$-\frac{n_1}{2} = -\frac{n_2}{2} \Rightarrow n_1 = n_2$

4 One odd, one even: one image is ≥ 0, the other < 0. ✓ ½
$f(n_1) \neq f(n_2)$, so this case cannot occur

5 Hence f is one-one.

6 Onto: let $y \in \mathbb{Z}$. If $y \geq 0$, take the odd number $n = 2y + 1$. ✓ ½
$f(2y+1) = \frac{2y+1-1}{2} = y$

7 If $y < 0$, take the even natural number $n = -2y$. ✓ ½
$f(-2y) = -\frac{-2y}{2} = y$

8 Every integer has a pre-image in $\mathbb{N}$. Hence f is one-one and onto. ✓ ½

Answer f **is both one-one and onto (bijective).**

Watch out: For $y < 0$ taking $n = 2y$, which is negative — use $n = -2y$, a natural number.

12 **Short answer** Board-pattern 3 marks

Show that the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = \dfrac{x^2}{1+x^2}$ is neither one-one nor onto. Further, find the set A so that $f : \mathbb{R} \to A$ becomes an onto function.

Formula / idea: Range of f: the set of y for which $f(x) = y$ has a real solution x.

1 Not one-one: compare $x = 1$ and $x = -1$. ✓ 1

$$f(1) = \frac{1}{2} = f(-1), \text{ but } 1 \neq -1$$

2 Not onto: $x^2 \geq 0$ and $1 + x^2 > 0$, so $f(x) \geq 0$. ✓ 1

$$-1 \in \mathbb{R} \text{ has no pre-image } \Rightarrow \ f \text{ is not onto}$$

3 Let $f(x) = y$ and solve for x^2.

$$x^2 = y + yx^2 \ \Rightarrow \ x^2(1-y) = y$$

4 $y = 1$ gives $0 = 1$, impossible. For $y \neq 1$:

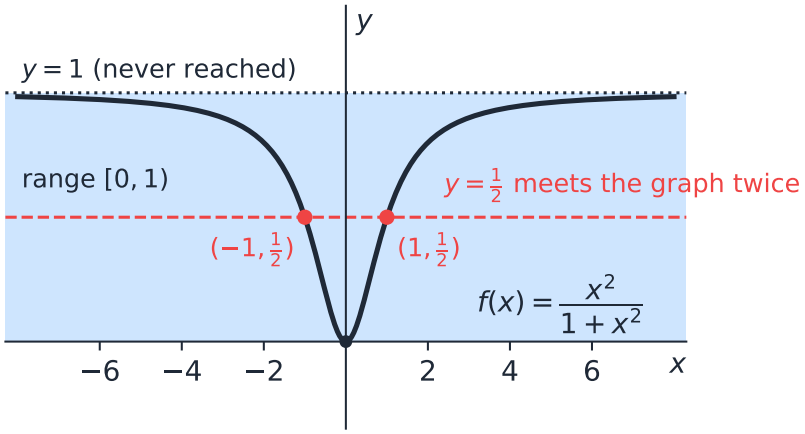
$$x^2 = \frac{y}{1-y}$$

5 Real x needs $x^2 \geq 0$.

$$\frac{y}{1-y} \geq 0 \ \Rightarrow \ 0 \leq y < 1$$

6 So the range of f is $[0, 1)$. Take A equal to it. ✓ 1

$$A = [0, 1)$$



Answer **Neither one-one nor onto; $A = [0, 1)$.**

Watch out: Writing $A = [0, 1]$ — $f(x) = 1$ needs $x^2 = 1 + x^2$, which is impossible.

13 **Short answer** PYQ-style 3 marks

Let $A = \mathbb{R} - \{-1\}$ and $f : A \to \mathbb{R} - \{b\}$ be defined by $f(x) = \dfrac{2x-3}{x+1}$. Find the value of b for which f is onto. For this b, show that f is also one-one.

Formula / idea: b is the one real number that $f(x)$ can never equal.

1 Let $f(x) = y$ and solve for x.
$2x - 3 = xy + y \;\Rightarrow\; x(2-y) = y + 3 \;\Rightarrow\; x = \dfrac{y+3}{2-y}$

2 This fails only at $y = 2$; check it directly.
$\dfrac{2x-3}{x+1} = 2 \;\Rightarrow\; 2x - 3 = 2x + 2 \;\Rightarrow\; -3 = 2$

3 Impossible, so 2 is never an image. Hence $b = 2$. ✓ 1

4 Onto: for $y \neq 2$, the pre-image $x = \dfrac{y+3}{2-y}$ is never -1.
$\dfrac{y+3}{2-y} = -1 \;\Rightarrow\; y + 3 = y - 2, \text{ impossible}$

5 Check $f(x) = y$ (multiply top and bottom by $2 - y$).
$f(x) = \dfrac{2(y+3) - 3(2-y)}{(y+3) + (2-y)} = \dfrac{5y}{5} = y$ ✓ 1

6 One-one: let $f(x_1) = f(x_2)$ and cross-multiply.
$(2x_1 - 3)(x_2 + 1) = (2x_2 - 3)(x_1 + 1)$ ✓ ½

7 Expand and cancel $2x_1x_2 - 3$.
$5x_1 = 5x_2 \;\Rightarrow\; x_1 = x_2$ ✓ ½

8 Hence with $b = 2$, f is one-one and onto.

Answer $b = 2$; f **is one-one and onto.**

Watch out: Not checking that the pre-image $\dfrac{y+3}{2-y}$ avoids -1, the excluded point of the domain.

14 **Short answer** NCERT Exemplar Ex 1.3 Q17 2 marks

Let R be the relation defined on the set of natural numbers $\mathbb{N}$ as $R = \{(x, y) : x \in \mathbb{N}, y \in \mathbb{N}, 2x + y = 41\}$. Find the domain and range of the relation R. Also verify whether R is reflexive, symmetric and transitive.

Formula / idea: $y = 41 - 2x$ must be a natural number, so $x \leq 20$.

1 $y = 41 - 2x \geq 1 \Rightarrow x \leq 20$. List domain and range.
$\text{Domain} = \{1, 2, \ldots, 20\}, \quad \text{Range} = \{39, 37, \ldots, 3, 1\}$ ✓ ½

2	Reflexive? $(1,1)$: $2(1)+1=3\neq 41 \Rightarrow (1,1)\notin R$	✓ ½
3	Symmetric? $(1,39)\in R$, but: $2(39)+1=79\neq 41 \Rightarrow (39,1)\notin R$	✓ ½
4	Transitive? $(11,19),(19,3)\in R$, but: $2(11)+3=25\neq 41 \Rightarrow (11,3)\notin R$	✓ ½
5	Hence R is neither reflexive, nor symmetric, nor transitive.	

Answer **Domain $=\{1,2,\ldots,20\}$, range $=\{1,3,5,\ldots,39\}$; R is neither reflexive, nor symmetric, nor transitive.**

Watch out: Taking x up to $41-x=21$ already gives $y=-1\notin\mathbb{N}$.

Level 3

25 marks

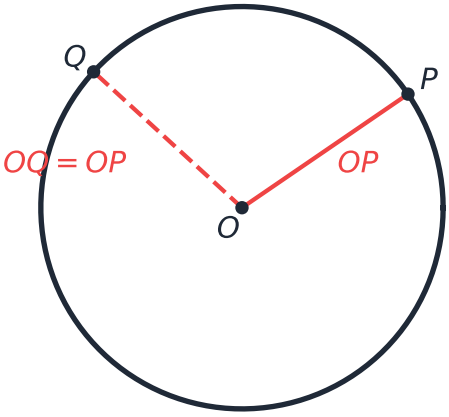
15 **Short answer** NCERT Ex 1.1 Q11 3 marks

Show that the relation R in the set A of points in a plane given by $R=\{(P,Q):\text{distance of the point } P \text{ from the origin is same as the distance of the point } Q \text{ from the origin}\}$, is an equivalence relation. Further, show that the set of all points related to a point $P\neq(0,0)$ is the circle passing through P with origin as centre.

Formula / idea: With O the origin, $(P,Q)\in R \iff OP=OQ$.

1	Reflexive: for every point P. $OP=OP \Rightarrow (P,P)\in R$	✓ ½
2	Symmetric: let $(P,Q)\in R$. $OP=OQ \Rightarrow OQ=OP \Rightarrow (Q,P)\in R$	✓ ½
3	Transitive: let $(P,Q),(Q,S)\in R$. $OP=OQ=OS \Rightarrow (P,S)\in R$	✓ ½
4	Hence R is an equivalence relation.	✓ ½
5	Points related to P: all Q with $OQ=OP$. $\{Q:OQ=OP\}$	

6	So these points form the circle with centre O and radius OP, passing through P.	✓ 1



circle: centre *O*, radius *OP*

Answer **R is an equivalence relation; points related to P form the circle with centre O and radius OP.**

Watch out: Describing the set as 'a circle' without naming its centre O and radius OP.

16 **Long answer** CBSE SQP 2026-27 Q32(A) 5 marks

Let $A = \mathbb{N} \times \mathbb{N}$, where $\mathbb{N}$ is the set of natural numbers and a relation R is defined on A such that $(a, b)\, R\, (c, d)$ if and only if $a^2 + d^2 = b^2 + c^2$. Check whether R is an equivalence relation or not.

Formula / idea: Rewrite the condition as $a^2 - b^2 = c^2 - d^2$.

1	Rearrange the defining condition. $(a, b)\, R\, (c, d) \iff a^2 - b^2 = c^2 - d^2$	✓ ½
2	Reflexive: for any $(a, b) \in A$. $a^2 + b^2 = b^2 + a^2 \Rightarrow (a, b)\, R\, (a, b)$	✓ 1
3	Symmetric: let $(a, b)\, R\, (c, d)$. $a^2 + d^2 = b^2 + c^2 \Rightarrow c^2 + b^2 = d^2 + a^2$	✓ ½
4	This is the condition for $(c, d)\, R\, (a, b)$. So R is symmetric.	✓ ½
5	Transitive: let $(a, b)\, R\, (c, d)$ and $(c, d)\, R\, (e, f)$. $a^2 - b^2 = c^2 - d^2 \quad \text{and} \quad c^2 - d^2 = e^2 - f^2$	
6	Equate the two results. $a^2 - b^2 = e^2 - f^2$	✓ 1
7	Rearrange. $a^2 + f^2 = b^2 + e^2 \Rightarrow (a, b)\, R\, (e, f)$	✓ 1

8 So R is transitive. Hence R is an equivalence relation. ✓ ½

Answer R **is reflexive, symmetric and transitive, hence an equivalence relation.**

Watch out: Writing the symmetric condition as $c^2 + d^2 = a^2 + b^2$ — for $(c, d)\, R\, (a, b)$ it is $c^2 + b^2 = d^2 + a^2$.

17 **Long answer** NCERT Exemplar Ex 1.3 Q21 5 marks

Let $A = [-1, 1]$. Discuss whether the following functions from A to A are one-one, onto or bijective:

(i) $f(x) = \dfrac{x}{2}$ 1 mark

(ii) $g(x) = |x|$ 1 mark

(iii) $h(x) = x|x|$ 2 marks

(iv) $k(x) = x^2$ 1 mark

Formula / idea: Onto $A \to A$ means the range must be the whole of $[-1, 1]$.

(i) 1 mark

1 One-one: $\dfrac{x_1}{2} = \dfrac{x_2}{2} \Rightarrow x_1 = x_2$. ✓ ½

2 Onto? For $x \in [-1, 1]$, $f(x) \in \left[-\dfrac{1}{2}, \dfrac{1}{2}\right]$. So $1 \in A$ has no pre-image. ✓ ½

$f(x) = 1 \;\Rightarrow\; x = 2 \notin A$

3 Hence f is one-one but not onto.

Answer (i) **One-one but not onto.**

(ii) 1 mark

1 Not one-one: ✓ ½

$g(-1) = g(1) = 1, \text{ but } -1 \neq 1$

2 Not onto: $|x| \geq 0$, so $-1 \in A$ has no pre-image. ✓ ½

3 Hence g is neither one-one nor onto.

Answer (ii) **Neither one-one nor onto.**

(iii) 2 marks

1 Remove the modulus.

$$h(x) = \begin{cases} x^2, & 0 \leq x \leq 1 \\ -x^2, & -1 \leq x < 0 \end{cases}$$

2 One-one: if x_1, x_2 have the same sign, $x_1^2 = x_2^2 \Rightarrow x_1 = x_2$. ✓ ½

3	If the signs differ, one image is ≥ 0 and the other < 0. $h(x_1) \neq h(x_2) \Rightarrow h$ is one-one	✓ ½
4	Onto: for $0 \leq y \leq 1$, take $x = \sqrt{y} \in A$. $h(\sqrt{y}) = (\sqrt{y})^2 = y$	
5	For $-1 \leq y < 0$, take $x = -\sqrt{-y} \in A$. $h(-\sqrt{-y}) = -\left(\sqrt{-y}\right)^2 = -(-y) = y$	✓ ½
6	Every $y \in A$ has a pre-image. Hence h is bijective.	✓ ½

Answer (iii) Bijective (one-one and onto).

(iv)

1 mark

1	Not one-one: $k(-1) = k(1) = 1, \text{ but } -1 \neq 1$	✓ ½
2	Not onto: $x^2 \geq 0$, so $-1 \in A$ has no pre-image.	✓ ½
3	Hence k is neither one-one nor onto.	

Answer (iv) Neither one-one nor onto.

Watch out: Calling $f(x) = \frac{x}{2}$ onto — it only reaches $\left[-\frac{1}{2}, \frac{1}{2}\right]$, not all of A.

18 Long answer Board-pattern

5 marks

Let $A = \{x \in \mathbb{Z} : 0 \leq x \leq 10\}$. Show that the relation $R = \{(a, b) : a, b \in A,\ a^2 - b^2 \text{ is divisible by } 5\}$ is an equivalence relation on A. Write the equivalence classes $[1]$ and $[2]$. How many distinct equivalence classes does R have?

Formula / idea: a and b are related exactly when a^2 and b^2 leave the same remainder on division by 5.

1	Reflexive: for every $a \in A$. $a^2 - a^2 = 0, \text{ divisible by } 5 \Rightarrow (a, a) \in R$	✓ 1
2	Symmetric: let $a^2 - b^2 = 5k$. $b^2 - a^2 = 5(-k) \Rightarrow (b, a) \in R$	✓ 1
3	Transitive: let $a^2 - b^2 = 5k$ and $b^2 - c^2 = 5m$. Add them. $a^2 - c^2 = 5(k + m) \Rightarrow (a, c) \in R$	✓ 1
4	Hence R is an equivalence relation.	
5	Remainders of a^2 on division by 5 for $a = 0, 1, \ldots, 10$: $0,\ 1,\ 4,\ 4,\ 1,\ 0,\ 1,\ 4,\ 4,\ 1,\ 0$	✓ ½
6	$[1]$: all b with remainder 1, like 1^2. $[1] = \{1, 4, 6, 9\}$	✓ ½

7	$[2]$: all b with remainder 4, like 2^2. $[2] = \{2, 3, 7, 8\}$	✓ ½
8	The rest form $[0] = \{0, 5, 10\}$. So R has 3 distinct equivalence classes.	✓ ½

Answer R **is an equivalence relation;** $[1] = \{1, 4, 6, 9\}$**,** $[2] = \{2, 3, 7, 8\}$**;** 3 **classes.**

Watch out: Putting 3 in $[1]$ — $3^2 - 1^2 = 8$ is not divisible by 5; check squares, not the numbers.

19 **Short answer** Board-pattern 3 marks

Show that the function $f : [1, 3] \to \mathbb{R}$ defined by $f(x) = x^2 - 2x + 3$ is one-one. Also find the range of f.

Formula / idea: $x^2 - 2x + 3 = (x - 1)^2 + 2$, and $x - 1 \geq 0$ on $[1, 3]$.

1	One-one: let $f(x_1) = f(x_2)$ for $x_1, x_2 \in [1, 3]$. $x_1^2 - x_2^2 - 2(x_1 - x_2) = 0 \Rightarrow (x_1 - x_2)(x_1 + x_2 - 2) = 0$	✓ ½
2	$x_1 + x_2 - 2 = 0$ only if $x_1 = x_2 = 1$, since both are ≥ 1. $\Rightarrow x_1 = x_2$ in every case	✓ 1
3	So f is one-one.	
4	Range: write $f(x) = (x - 1)^2 + 2$ with $0 \leq x - 1 \leq 2$. $0 \leq (x - 1)^2 \leq 4 \Rightarrow 2 \leq f(x) \leq 6$	✓ ½
5	Each $y \in [2, 6]$ is reached at $x = 1 + \sqrt{y - 2} \in [1, 3]$. $f\left(1 + \sqrt{y - 2}\right) = (y - 2) + 2 = y$	✓ ½
6	Hence the range of f is $[2, 6]$.	✓ ½

Answer f **is one-one; range of** $f = [2, 6]$**.**

Watch out: Writing the range as $[f(1), f(3)]$ without showing $2 \leq f(x) \leq 6$ for every $x \in [1, 3]$.

20 **Case study** CBSE SQP 2024-25 Q37 4 marks

Case Study: An organization conducted bike race under 2 different categories — boys and girls. In all, there were 250 participants. Among all of them finally three from Category 1 and two from Category 2 were selected for the final race. Ravi forms two sets B and G with these participants for his college project. Let $B = \{b_1, b_2, b_3\}$, $G = \{g_1, g_2\}$, where B represents the set of boys selected and G the set of girls who were selected for the final race. Ravi decides to explore these sets for various types of relations and functions.

(i) Ravi wishes to form all the relations possible from B to G. How many such relations are possible? 1 mark

(ii) Write the smallest equivalence relation on G. 1 mark

(iii)(a) Ravi defines a relation from B to B as $R_1 = \{(b_1, b_2), (b_2, b_1)\}$. Write the minimum ordered pairs to be added in R_1 so that it becomes (A) reflexive but not symmetric, (B) reflexive and symmetric but not transitive. 2 marks

OR

(iii)(b) If the track of the final race (for the biker b_1) follows the curve $x^2 = 4y$ (where $0 \le x \le 20\sqrt{2}$ and $0 \le y \le 200$), then state whether the track represents a one-one and onto function or not. (Justify) 2 marks

Formula / idea: A relation from B to G is any subset of $B \times G$.

(i) 1 mark

1 Count the elements of $B \times G$.
$n(B \times G) = 3 \times 2 = 6$

2 Each relation is a subset of $B \times G$.
$\text{Number of relations} = 2^6 = 64$ ✓ 1

Answer (i) 64 **relations.**

(ii) 1 mark

1 An equivalence relation must contain every (g, g); these alone are already symmetric and transitive.

2 Smallest equivalence relation on G:
$\{(g_1, g_1), (g_2, g_2)\}$ ✓ 1

Answer (ii) $\{(g_1, g_1), (g_2, g_2)\}$

(iii)(a) 2 marks

1 (A) Reflexive needs $(b_1, b_1), (b_2, b_2), (b_3, b_3)$. The result is still symmetric, so add one pair without its reverse.

2 Minimum pairs to add for (A): ✓ 1

$(b_1, b_1), (b_2, b_2), (b_3, b_3), (b_2, b_3)$

(b_3, b_2), (b_1, b_3) or (b_3, b_1) also works for (b_2, b_3)

3 (B) R_1 with the three (b, b) is transitive; add a pair and its reverse to break transitivity.

$(b_1, b_2), (b_2, b_3) \in R$ but $(b_1, b_3) \notin R$

4 Minimum pairs to add for (B): ✓ 1

$(b_1, b_1), (b_2, b_2), (b_3, b_3), (b_2, b_3), (b_3, b_2)$

$(b_1, b_3), (b_3, b_1)$ in place of the last two also works

Answer (iii)(a) **(A) add $(b_1, b_1), (b_2, b_2), (b_3, b_3), (b_2, b_3)$; (B) add $(b_1, b_1), (b_2, b_2), (b_3, b_3), (b_2, b_3), (b_3, b_2)$ (other choices noted above are also correct)**

OR

(iii)(b)

2 marks

1 Write the track as a function.

$f : [0, 20\sqrt{2}] \to [0, 200], \quad f(x) = \dfrac{x^2}{4}$

2 One-one: let $f(x_1) = f(x_2)$. ✓ 1

$x_1^2 = x_2^2 \Rightarrow (x_1 - x_2)(x_1 + x_2) = 0 \Rightarrow x_1 = x_2$

$x_1, x_2 \geq 0$, so $x_1 + x_2 = 0$ only if both are 0

3 Onto: for $y \in [0, 200]$ take $x = 2\sqrt{y} \in [0, 20\sqrt{2}]$. ✓ 1

$f(2\sqrt{y}) = \dfrac{4y}{4} = y$

4 Hence the track represents a one-one and onto function.

Answer (iii)(b) **Yes — it is a one-one and onto function.**

Watch out: Answering 6 in (i) — that is $n(B \times G)$; the relations are its subsets, $2^6 = 64$.